Introduction to Large Language Models
Assignment- 4

TYPE OF QUESTION: MCQ
Number of questions: 9 **Total mark: 8 X 1 + 1 X 2 = 10**

QUESTION 1: [1 mark]

State True or False.

In the Continuous Bag of Words (CBOW) model, the surrounding context words are used to predict the centre target word.

a. True
b. False

Correct Answer: a

Solution: CBOW aggregates the context window and predicts the centre word, while Skip-gram does the reverse - it uses the centre word to predict each surrounding context word.

QUESTION 2: [1 mark]

Two distinct words in a vocabulary are represented as one-hot vectors. What is the cosine similarity between them?

a. 1
b. 0
c. -1
d. It depends on the size of the vocabulary

Correct Answer: b

Solution: One-hot vectors for distinct words have their non-zero entries at different indices, so their dot product is 0. The cosine similarity is therefore 0 for every pair of distinct words, which is exactly why one-hot encoding carries no notion of semantic similarity.

QUESTION 3: [1 mark]

A toy corpus has an initial base vocabulary of 8 unique characters. If the Byte Pair Encoding (BPE) algorithm performs exactly 5 merge operations, what is the final size of the vocabulary?

a. 8
b. 5
c. 13
d. 40

Correct Answer: c

Solution: Each merge operation adds exactly one new token to the vocabulary and the base characters are retained, so the final size is 8 + 5 = 13.

QUESTION 4: [1 mark]

Which statement correctly describes how WordPiece chooses the next pair to merge, compared with BPE?

a. BPE merges the most frequent pair, while WordPiece merges the pair maximising Frequency(A, B) / (Frequency(A) x Frequency(B))
b. BPE merges the least frequent pair, while WordPiece merges the most frequent pair

c. Both merge the pair with the highest raw co-occurrence frequency

d. WordPiece merges the pair maximising Frequency(A) x Frequency(B) / Frequency(A, B)

Correct Answer: a

Solution: BPE is purely frequency-driven. WordPiece normalises the pair frequency by the frequencies of the individual tokens, so a pair is merged only when the two pieces co-occur more than their individual frequencies would suggest.

QUESTION 5: [1 mark]

Which of the following statements about subword tokenization are true?

a. It guarantees that every word in the test set is present in the vocabulary as a single token.

b. Rare and morphologically complex words are split into meaningful subword units.

c. It offers a middle ground between word-level and character-level tokenization.

d. It keeps the vocabulary size bounded while still being able to represent unseen words.

Correct Answer: b, c, d

Solution: Option (a) is false - subword tokenizers explicitly allow a word to be represented by a sequence of several tokens; that is how they handle unseen words with a bounded vocabulary.

QUESTION 6: [2 marks]

Consider the following word-context count matrix:

	c1	c2	c3	c4	c5
w1	2	0	3	1	4
w2	1	3	0	2	2
w3	4	1	2	0	3

Compute the cosine similarity between w1 and w2.

a. 0.516

b. 0.867

c. 0.400

d. 0.712

Correct Answer: a

Solution: Dot product: (2)(1) + (0)(3) + (3)(0) + (1)(2) + (4)(2) = 2 + 0 + 0 + 2 + 8 = 12

||w1|| = sqrt(4 + 0 + 9 + 1 + 16) = sqrt(30) = 5.4772

||w2|| = sqrt(1 + 9 + 0 + 4 + 4) = sqrt(18) = 4.2426

cos(w1, w2) = 12 / (5.4772 x 4.2426) = 12 / 23.238 = 0.516

QUESTION 7: [1 mark]

Why is Word2Vec training described as self-supervised?

a. The training labels are produced by human annotators before training begins.

b. The corpus itself supplies the labels, since the context of a word acts as its supervision signal.

c. The model supervises its own learning rate schedule during training.

d. An external labelled dataset is required to validate every update.

Correct Answer: b

Solution: No manual annotation is needed. Positive training pairs are read directly off the raw text using a sliding context window, so the data generates its own supervision.

QUESTION 8: [1 mark]

What is the purpose of negative sampling in Word2Vec training?

a. To reduce the dimensionality of the learned word vectors
b. To guarantee that gradient descent converges to a global optimum
c. To avoid computing the softmax normalisation over the entire vocabulary
d. To remove stop words from the training corpus

Correct Answer: c

Solution: The full softmax requires a sum over every word in the vocabulary at each update. Negative sampling replaces this with a small number of binary logistic decisions against randomly drawn negative words, making each update cheap and independent of vocabulary size.

QUESTION 9: [1 mark]

Which of the following representations handles Out-Of-Vocabulary (OOV) words best?

a. One-hot encoding
b. Count-based SVD embeddings
c. CBOW Word2Vec
d. FastText embeddings

Correct Answer: d

Solution: FastText represents a word as the sum of its character n-gram vectors, so an embedding can be composed for a word never seen during training. The other three assign vectors only to words present in the training vocabulary.
